

Kinetic Light Sculpture

Authorize one bounded motif; mirror every logical step

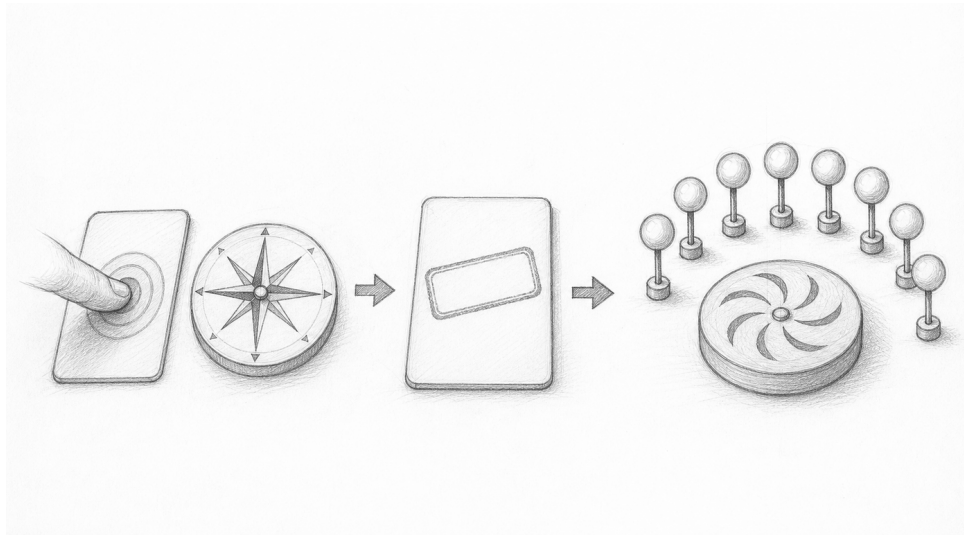
E0 COPIED REPLAY — NO POWERED INPUT, LIGHT, OR MOTOR CLAIM

Lesson 048 — project
Prerequisites Lessons 046–047
Evidence named memory result cells

Time 140–200 minutes
Board compile-only Mega replay
Bounds logical position $[-256, 256]$

Project boundary. This workbook composes copied tactile and directional evidence with a pure logical step sequence and copied light intent. It does not read a pin, energize an LED or coil, move a shaft, prove position, qualify a driver, or provide a wiring plan.

1 Stage authorization before motion



Alternative text: a graphite-pencil sequence separates a finger touch and eight-way direction, a retained pending authorization card, and an inert imagined rotor beside eight lamps. Motion never begins in the same frame that creates authorization.

A valid touch event plus current directional evidence creates one **Pending** authorization, even when its direction is Neutral. That originating frame consumes no step and publishes no coil pattern. A healthy, non-neutral, strictly forward frame may accept it as motion; Neutral, stop, fault, expiry, or shutdown resolves it with the corresponding non-motion outcome. For acceptance, the coordinator chooses one of eight fixed motifs, asks the Lesson 047 sequencer to prepare it, verifies both prepared candidates, and commits the transaction.

1. **Acquire copied evidence:** one contact source, one directional source, one independent stop source, and one project frame identity.
2. **Preview without mutation:** validate time, source domain, replay identity, age, skew, quality, and bounds.
3. **Authorize once:** retain source identities, sequences, direction, originating frame, and **Pending** disposition.
4. **Resolve later:** on the next healthy forward frame, accept one bounded non-neutral motif; Neutral resolves inhibited, while bounds, stop, fault, or expiry preserve their distinct terminal outcome.

The resolving frame exposes the terminal authorization as **Accepted**, **Inhibited**, or **BoundRejected**. The next accepted forward frame clears that terminal record unless it replaces it with a new resolution. Exact replay never authorizes twice.

2 Map eight directions to fixed motifs

Direction	logical direction	steps	interval
North	forward	8	120 ms
NorthEast	forward	12	100 ms
East	forward	16	80 ms
SouthEast	forward	12	100 ms
South	reverse	8	120 ms
SouthWest	reverse	12	100 ms
West	reverse	16	80 ms
NorthWest	reverse	12	100 ms

These are project motifs, not physical angles or distances. The logical position must remain inside $[-256, 256]$. A request crossing either bound becomes **BoundRejected**: no motion starts, no step is consumed, and the travel-limit result cell becomes true. A fresh opposite authorization can recover.

3 Mirror coil intent exactly

On every accepted frame,

```
lights.shiftRegisterBits = motion.coilIntent.
```

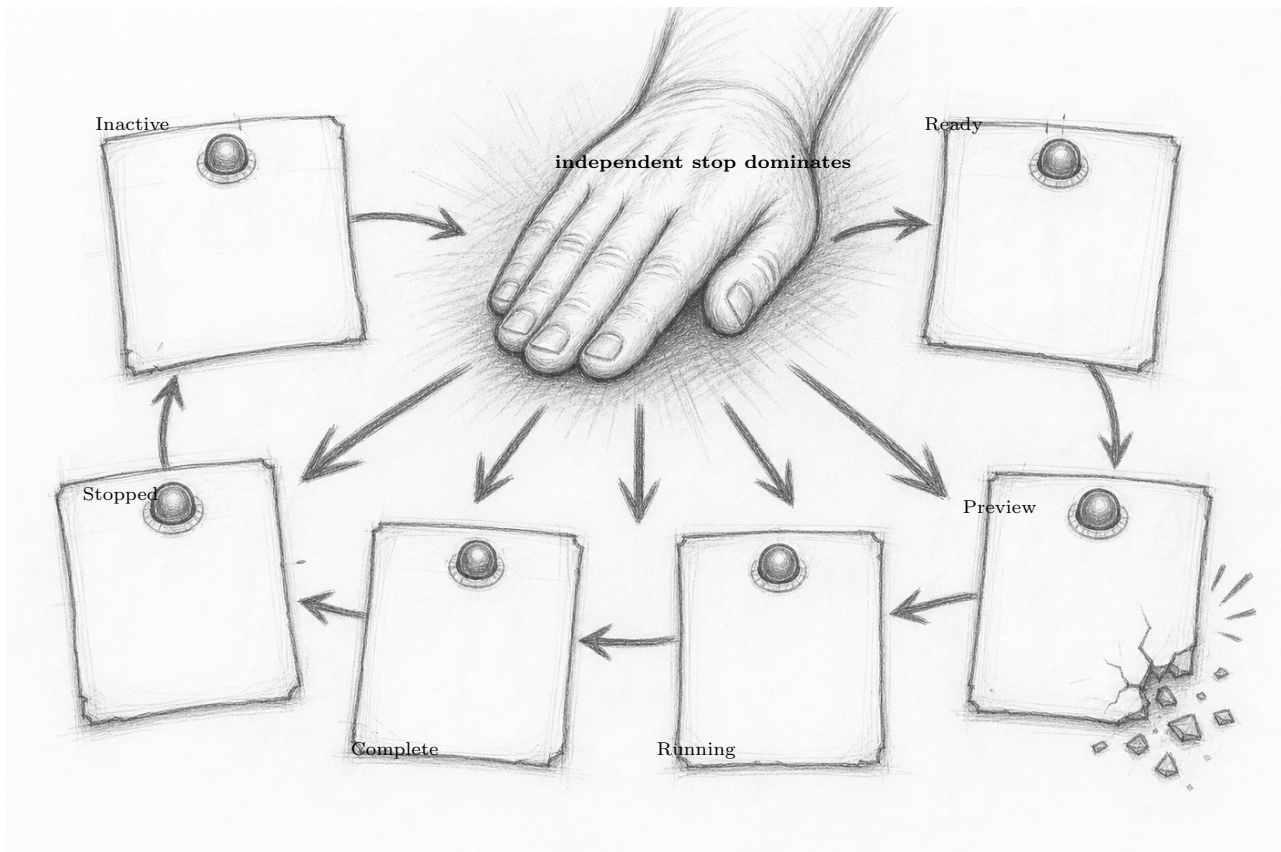
The equality is exact, including zero during authorization, cancellation, stop, fault, expiry, and shutdown. The light record is copied semantic intent; it is neither a shift-register driver nor evidence that a lamp was acquired.

frame	1	2	3	4	5	6	7	8
coil byte	01	03	02	06	04	0C	08	09
light byte	01	03	02	06	04	0C	08	09

The row is one complete eight-frame logical sequence, not the direction-to- motif table above. Reverse motion traverses the same configured frame order in reverse. Because upper coil-intent bits are zero, upper light-byte bits are also zero.

frame ID	coil intent	light bits	equal?
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

4 Let stop dominate the transaction



Alternative text: a rough pencil state loop contains inactive, ready, preview, running, complete, and stopped cards, while a large hand is a metaphor for copied independent-stop precedence. Fault is described in the adjacent table; this is not a physical control layout.

A valid active stop is assessed before malformed interaction evidence. It cancels prepared or running motion, clears coil and light bits, inhibits a pending authorization, and latches **Stopped**. A held stop cannot restart. Release permits later healthy evidence to return to **Ready**; release itself never creates authorization.

Outcome	Required project result	Motion/light result
Ready	healthy copied evidence; no pending motion	bits zero
Preview	pending authorization retained	steps and bits zero
Running	terminal Accepted on resolving frame	exact mirror
Complete	requested logical steps consumed	hold/off per policy
Stopped	valid independent stop dominates	coils and mirror zero
Fault	admitted source/motion failure latched	coils and mirror zero

5 Distinguish rejection, stop, and fault

Malformed frame identity, future time, ambiguous half-range order, changed same-ID payload, impossible copied values, or unrecognized enum values reject atomically. The prior project snapshot does not partially advance.

An admitted stop is not a fault. An admitted producer or sequence failure is a fault and publishes a complete fault record with zero coil/light intent. Fault is latched for the object lifetime; later healthy evidence and replay cannot clear it. Recovery requires **shutdown()** followed by successful **initialize()**. Shutdown returns **Inactive**, cancels motion, publishes all-zero intent, and terminalizes a pending authorization as **Inhibited**. That terminal record remains inspectable until successful initialization clears the epoch.

Stop semantics at E0. The copied stop record proves policy precedence only. It is not an emergency-stop component, a safety-rated function, or proof that physical power was independently removed.

6 Run the deterministic E0 workbook

1. Initialize. Predict **Ready**, logical position zero, no authorization, and all intent bits zero.
2. Qualify a touch while supplying North. Predict **Pending** and zero requested steps in that frame. Repeat with Neutral: it still becomes pending, then resolves **Inhibited**.
3. Supply a healthy forward frame. Predict **Accepted**, forward eight-step motion, 120 ms cadence, and exact coil/light equality.
4. Repeat for all eight directions. Record the chosen direction, steps, cadence, and motif without claiming shaft motion or emitted light.
5. Replay the resolving frame exactly. Predict no second motif. Supply one accepted forward frame and verify the terminal record clears.
6. Collide an active valid stop with malformed directional evidence. Predict **Stopped**, pending disposition **Inhibited**, and all-zero intent.
7. Drive copied logical position toward 256, then request a motif that would exceed it. Predict a bound rejection, travel-limit intent, and no step. Repeat at -256 .
8. Inject source failure, command expiry, timing rollover, and shutdown. Predict fault-safe or inactive all-zero intent without partial state.

E0 result cells

frame	phase	authorization	position	status
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
direction	steps	coil bits	light bits	stop/limit
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____

7 Interpret only what E0 observes

Acquisition prediction. Recognized nonzero source identities, forward-or-repeat chronology, bounded age/skew, and self-consistent copied contact, direction, and stop records form one admissible frame.

Acquisition observation. Inspect **frameId**, copied source identities, sequences, ages, qualities, project status, and accepted authorization records in host memory.

Acquisition interpretation. Acceptance proves the pure copied-data contract. It does not prove a touch board, joystick, stop switch, clock, pin, or electrical endpoint was acquired.

Safe-state prediction. Stop, fault, expiry, bound rejection, and shutdown publish the specified terminal disposition and zero motion/light intent where required.

Safe-state observation. Inspect phase, motion disposition, coil intent, light byte, pending/terminal records, travel-limit flag, and status.

Safe-state interpretation. An all-zero memory record proves only the E0 policy result. It does not prove de-energized coils, dark LEDs, stationary hardware, or removed motor power.

8 Bound the compile-only composition

The canonical Mega replay uses fixed storage and no heap, live input GPIO, stepper driver, shift-register endpoint, motor supply, timer-owned motion, or hardware debugger. Its final measured package size is:

Artifact	measured	budget
Mega flash	22,216 bytes	28,672 bytes
static SRAM	1,470 bytes	2,048 bytes
KineticLightSculpture object	502 bytes	512 / 640 bytes
static compiler/link stack plus timer-0 ISR	687 bytes	evidence only
SRAM remaining after static and stack/ISR	6,035 bytes	1,024-byte margin

The 687-byte estimate is conservative static compiler/link evidence, not runtime high-water. Compilation is packaging evidence, not runtime stack, current, timing, temperature, motion, or electrical shutdown evidence.

9 Leave E1 input and light acceptance blank

E1 remains open. E1 may qualify exact low-energy tactile, directional, stop, and indicator specimens. It still may not energize a stepper coil or moving load.

Exact specimen and revision markings	_____
Primary datasheet sources	_____
Supply, logic levels, and pin order	_____
Input qualification and active polarity	_____
Indicator resistors and measured current	_____
Independent stop observation	_____
Electrical rollback observation	_____
Reviewer, date, acceptance record	_____

10 Leave E2 motion acceptance blank

E2 remains open. A likely 28BYJ-48 label and ULN2003 carrier do not establish motor voltage, winding, driver topology, clamp connection, safe current, or motion. No physical build is authorized here.

Exact motor, winding, and driver identity	_____
Electrically authoritative schematic review	_____
Separate current-limited supply and common	_____
Measured coil and aggregate current	_____
Thermal, cadence, and restraint evidence	_____
Measured travel and bound behavior	_____
Independent power-removal safe state	_____
Reviewer, date, acceptance record	_____

Future E2 work must use a lightweight, stable tabletop mechanism with guarded pinch points, no loose load, no sharp or high-inertia feature, no stored energy, and a stop/power-removal path independent of software. This project never controls a launcher, vehicle, door lock, lifting mechanism, or safety device.

11 Software evidence and further work

Deterministic tests cover initialization, all eight motifs, staged and replayed authorization, terminal-record lifetime, transaction rollback, source-domain changes, stop dominance and release, expiry, bounds, fault reinitialization, rollover, storage lifetime, light equality, and shutdown.

- Canonical compile-only Lesson 048 sketch.
- Deterministic kinetic-sculpture tests.
- Kinetic-sculpture interface.
- Lessons 046–048 design record.
- Deterministic testing contract.
- Safety model.

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